

FLL Rulebook: Easy Read Breakdown

Welcome & Season Overview

- **Summary:** This season's theme is **UNEARTHED™**—an archaeology adventure where your robot explores a dig site, carefully uncovers artifacts, and tells their stories.
- **Key Points:**
 - Events score four parts equally (25% each): **Robot Design, Innovation Project, Robot Performance, and Core Values.**
 - You'll run up to **three 2.5-minute matches**; only your **best** match score counts.
- **Tips for Kids:**
 - Think like a scientist: plan, test, learn, improve.
 - Core Values (teamwork, respect, fairness) matter everywhere—not just at the table.
- **Examples:**
 - If your first match is messy, fix code/attachments and try again—your best score is the one that counts.
- **Resources for Kids:**
 - Season overview and scoring rubrics (look for the **Game & Season Resources** page).

The Robot Game (What You Do in Matches)

- **Summary:** In each **2.5-minute** match, your autonomous LEGO® robot tries missions around the field to earn points.
- **Key Points:**
 - **Mission Strategy:** Pick which missions to try and in what order.
 - **Design & Code:** Build a reliable robot and write programs that run without help.
 - **Launch from Home:** Every run starts inside a **launch area** in **home**.
 - **Score at End:** Most mission results must be **visible at the end** to count.
- **Tips for Kids:**
 - Aim for **consistent** missions before chasing hard ones.
 - Use simple, sturdy attachments; test them a lot.
- **Examples:**
 - Program A: go to Mission 1, return to home; Program B: do Mission 2; chain them during the match.
- **Resources for Kids:**
 - Robot Game **Missions Video** and **Official Scoring Calculator**.

Inside the Challenge Set (What's in the Box)

- **Summary:** Your set includes the **mat**, **mission models**, and **Dual Lock** fasteners—build models exactly as shown so your practice matches tournaments.
- **Key Points:**
 - Mission models arrive in **numbered bags**; use the official **building instructions**.
 - Incorrect builds can change how missions behave.
- **Tips for Kids:**
 - Build in pairs: one reads, one builds; then swap to check.
- **Examples:**
 - A model that's not aligned can block your robot even if your code is perfect.
- **Resources for Kids:**
 - **Mission Model Building Instructions** (per bag list on the setup page).

Getting Started & Helpful Resources

- **Summary:** Decide where to put the mat (table or floor), build the table if you want, set up the field, then learn the missions and rules.
- **Key Points:**
 - Watch the **Field Setup Video** and **Missions Video**.
 - Check **Challenge Updates** often (official changes/clarifications).
 - Practice scoring with the **Official Scoring Calculator**.
 - **LEGO® Education SPIKE™ App** offers a guided mission to start coding.
- **Tips for Kids:**
 - Keep a **team notebook** to remember what worked and what didn't.
- **Examples:**
 - If your mat slides, use thin **matte gaffer's tape** on the edges.
- **Resources for Kids:**
 - Field Setup Video, Missions Video, Challenge Updates, Scoring Calculator, SPIKE App.

Field Setup (How to Place Everything)

- **Summary:** Roll out the mat, attach Dual Lock to marked spots, and place each model exactly on its wireframe.
- **Key Points:**
 - Press models down until you feel a **“click”** (Dual Lock engaged).
 - Before each match, double-check all start positions.
- **Tips for Kids:**
 - Make a **pre-match checklist** (models placed, robot batteries, attachments ready).
- **Examples:**
 - If a model seems off, ask the **referee** to check **before** the match starts.
- **Resources for Kids:**
 - Field setup steps and bag-to-mission table (in the guide).

Missions – How They Work (Read This First)

- **Summary:** Missions are tasks around the mat. You choose which to try. Results usually must be **visible at the end** to score.
- **Key Points:**
 - **Equipment Inspection bonus:** If all your equipment fits in **one** launch area under **12 in (305 mm)** at inspection, earn **+20 points**.
 - **“No Equipment Constraint” symbol:** If shown on a mission, that model **can’t be touching your equipment at the end** or it won’t score (only for missions that show the symbol).
 - **Precision Tokens:** Start with **6**; lose one if you interrupt your robot outside home; points depend on how many are left at the end (see section below).
- **Tips for Kids:**
 - Plan **short, reliable runs** that return to home, so you can swap attachments quickly.
- **Examples:**
 - Two small missions that always work can beat one risky big mission.
- **Resources for Kids:**
 - Missions pages and icons legend in the rulebook.

Mission 01: Surface Brushing

- **Summary:** Carefully brush away soil to uncover the site without hurting artifacts.
- **Key Points:**
 - **Soil deposits** completely cleared (touching the mat): **10 points each**.
 - **Archaeologist’s brush** not touching the dig site: **10 points**.
- **Tips for Kids:**
 - Use gentle contact—low power and a rounded attachment.
- **Examples:**
 - A hinged “sweeper” that glides over soil and returns to home.
- **Resources for Kids:** Missions Video (watch the brushing demo).

Mission 02: Mineshaft Explorer

- **Summary:** Navigate the mine safely and handle the minecarts.
- **Key Points:**
 - **Support structures** both standing: **10**.
 - Your **minecart** on the **opposing field**: **30**.
 - **Bonus:** and their minecart on **your** field: **+10**.
 - To score, your minecart must pass **completely through** the **mineshaft entry**.
- **Tips for Kids:**
 - Aim straight; use a **line follow** or guide rail on the mat.
- **Examples:**
 - A forked pusher that nudges the cart across the boundary.
- **Resources for Kids:** Mission card notes on the minecart path.

Mission 03: Map Reveal

- **Summary:** Clear **topsoil sections** to uncover a hidden map.
- **Key Points:**
 - **Topsoil sections** completely cleared: **10 points each**.
- **Tips for Kids:**
 - A wide, flat “bulldozer” attachment can clear more in one pass.
- **Examples:**
 - Drive along the boundary to keep direction steady.
- **Resources for Kids:** Mission diagram with topsoil locations.

Mission 04: Careful Recovery

- **Summary:** Remove a **precious artifact** without damaging the mine.
- **Key Points:**
 - **Precious artifact** not touching the mine: **30**.
- **Tips for Kids:**
 - Use a claw that **lifts up** before moving sideways.
- **Examples:**
 - A simple hook that raises, then backs away.
- **Resources for Kids:** Artifact model callouts.

Mission 05: Who Lived Here?

- **Summary:** Rebuild part of an old structure.
- **Key Points:**
 - **Structure floor** completely upright: **30**.
- **Tips for Kids:**
 - Approach squarely so you don’t knock it over again.
- **Examples:**
 - A vertical lifter that flips the floor up, then releases.
- **Resources for Kids:** Structure model layout.

Mission 06: Forge

- **Summary:** Release ore blocks used for making tools; one hides a fossilized artifact.
- **Key Points:**
 - **Ore blocks** not touching the forge: **10 each**.
 - Tech note: You may **open ore blocks by hand only when completely in home** (useful for Mission 13 “Forum”).
- **Tips for Kids:**
 - If two blocks are close, design an attachment that moves both at once.
- **Examples:**
 - A fingered arm that pulls, then sweeps ore away.
- **Resources for Kids:** Forge mechanism animation in the video.

Mission 07: Heavy Lifting

- **Summary:** Move a heavy **millstone**.
- **Key Points:**
 - **Millstone** no longer touching its base: **30**.
- **Tips for Kids:**
 - Use a wedge to **slide** it first, then a pusher to finish.
- **Examples:**
 - Low-friction skids so the stone doesn't bounce back.
- **Resources for Kids:** Millstone/base visuals.

Mission 08: Silo

- **Summary:** Remove preserved pieces from the silo for study.
- **Key Points:**
 - **Preserved pieces** outside the silo: **10 each**.
- **Tips for Kids:**
 - Collect pieces in a basket so they don't scatter.
- **Examples:**
 - Use the robot body as a "wall" to block roll-aways.
- **Resources for Kids:** Silo interior diagram.

Mission 09: What's on Sale?

- **Summary:** Restore the **market stall**.
- **Key Points:**
 - **Roof** completely raised: **20**.
 - **Market wares** raised: **10**.
- **Tips for Kids:**
 - Lift the roof **before** you push the wares to avoid jams.
- **Examples:**
 - A two-stage arm: first roof, then wares.
- **Resources for Kids:** Market model with roof hinge.

Mission 10: Angler Artifacts

- **Summary:** Use the crane at the port to lift delicate artifacts.
- **Key Points:**
 - **Artifacts raised** above ground layer: **20**.
 - **Bonus:** crane **flag** at least partly lowered: **+10**.
- **Tips for Kids:**
 - Approach slowly; small nudges beat hard hits.
- **Examples:**
 - A hook that catches the crane's loop and eases it down.
- **Resources for Kids:** Port crane close-ups.

Mission 11: Tip the Scales

- **Summary:** Use an ancient **scale** to show fair weighing.
- **Key Points:**
 - **Scale** tipped and **touching the mat**: **20**.
 - **Scale pan** completely removed: **10**.
- **Tips for Kids:**
 - Push near the **end** of a lever—more effect with less force.
- **Examples:**
 - A rounded bumper that tips without snagging.
- **Resources for Kids:** Scale lever diagram.

Mission 12: Salvage Operation

- **Summary:** Excavate a fragile **ship** buried in **sand**.
- **Key Points:**
 - **Sand** completely cleared: **20** (counts when the **pull activator** is past the line on the mat).
 - **Ship** completely raised: **10**.
- **Tips for Kids:**
 - Pull in a straight line so nothing twists or breaks.
- **Examples:**
 - A latch that grabs the pull activator, then reverses smoothly.
- **Resources for Kids:** Sand/ship mechanism notes.

Mission 13: Forum (Display Artifacts)

- **Summary:** Deliver selected artifacts to the **forum** area to share the story of the site.
- **Key Points:**
 - **Artifacts touching the mat and at least partly in the forum: 5 points each.** Eligible items:
 - **Brush, Topsoil, Precious Artifact, Opposing Team's Minecart, Ore with Fossilized Artifact, Millstone, Scale Pan.**
- **Tips for Kids:**
 - Plan a **return-to-home** route so you can bring items back safely before the last delivery.
- **Examples:**
 - A scoop that carries multiple pieces without dropping them.
- **Resources for Kids:** Forum zone outline on the mat.

Mission 14: Statue Rebuild

- **Summary:** Rebuild and raise an important **statue**.
- **Key Points:**
 - **Statue** completely raised: **30**.
- **Tips for Kids:**
 - Use alignment guides (edge of mat graphics) to approach straight.
- **Examples:**
 - A lifter that slides under and raises evenly.
- **Resources for Kids:** Statue mechanism in the video.

Mission 15: Site Marking

- **Summary:** Mark places to study later with **flags**.
- **Key Points:**
 - **Sites with a flag at least partly inside and touching the mat: 10 points each.** (Sites are outlined on the mat.)
- **Tips for Kids:**
 - Practice precise stops so flags land exactly in the outlined areas.
- **Examples:**
 - A spring-loaded dropper that releases a flag when you hit a line cue.
- **Resources for Kids:** Site outlines on the mat wireframe.

Rules – Glossary (Words You’ll Hear)

- **Summary:** Key words used at the table.
- **Key Points:**
 - **Match:** 2.5 minutes of autonomous robot play.
 - **Field:** Mat + models + border walls + home areas.
 - **Mission / Mission Model:** Tasks and the LEGO® models you interact with.
 - **Launch:** Start the robot **from rest** completely within a **launch area**.
 - **Interruption:** Any time a technician touches the robot after launch.
 - **Autonomous:** Robot runs **without help**.
- **Tips for Kids:**
 - Use **pseudocode** (plain steps) before coding.
- **Examples:**
 - Pseudocode: “1) drive to line, 2) follow line to model, 3) lower arm, 4) back to home.”
- **Resources for Kids:** Glossary page in the rulebook.

Rules – Before the Match: Equipment

- **Summary:** What you’re allowed to bring and use.
- **Key Points:**
 - **LEGO® parts only** (original condition). **String** and **pneumatic tubes** may be cut.
 - **Controllers:** 1 per match (SPIKE™ Prime/Essential, EV3, Robot Inventor, Powered Up, NXT, RCX allowed).
 - **Motors:** Up to **4** total (any mix).
 - **Sensors:** Touch/force, color, distance/ultrasonic, gyro (**any mix/number**).
 - **Power:** One controller power pack **or** six AA batteries; LEGO wires; one **microSD** card.
 - **Software:** Any language, but robot must be **autonomous outside home** (no remote control).
 - One **sheet of paper** per home area for notes (doesn’t count as equipment).
 - No extra or duplicate **mission models**.
- **Tips for Kids:**
 - Label attachments so you can swap them fast at home.
- **Examples:**
 - “Red Arm” for pushing, “Blue Scoop” for lifting.
- **Resources for Kids:** Equipment rules page.

Rules – Before the Match: Match Setup & Inspection

- **Summary:** How to set up at the table and pass inspection.
- **Key Points:**
 - All equipment must fit inside the **two launch areas** under **12 in (305 mm)**.
 - **Bonus +20 points** if everything fits in **one** launch area under 12 in.
 - No extra storage; only table edges (left/right) may hold gear (about **6.75 in x 45 in** each side).
 - Calibrate sensors on the mat and ask the referee to check anything **before** the match starts.
 - Team stands on **both sides** of the table; at most **2 technicians per home area** at a time.
- **Tips for Kids:**
 - Pre-sort attachments into **left** and **right** home to match your planned launches.
- **Examples:**
 - Left side: missions 1–3, Right side: missions 4–5.
- **Resources for Kids:** Pre-match checklist in the rulebook.

Rules – During the Match: Inside Home

- **Summary:** What technicians may do **inside** home.
- **Key Points:**
 - You can handle the **robot/equipment/models** only when they're **completely** inside your home area.
 - You **cannot** pass items from one home area to the other.
 - You **cannot** touch anything outside your home area (except to **interrupt** the robot).
- **Tips for Kids:**
 - Keep hands behind an imaginary line—home only.
- **Examples:**
 - If a wheel is halfway out of home, wait or use the robot to bring it in before touching.
- **Resources for Kids:** Inside-home do's and don'ts.

Rules – During the Match: Outside Home

- **Summary:** What happens when you interrupt the robot or handle items outside home.
- **Key Points:**
 - Try to let the robot **return completely to home** before interrupting.
 - If you interrupt while the robot (or what it's touching) is **outside home**, you **lose one precision token** and must **relaunch**.
 - **Partly outside:** Return the robot/items to the **same** home area.
 - **Completely outside:** You may place them in **either** home area.
 - Items obtained **outside home** after the last launch must be **given to the referee** if you interrupt.
 - You may **stop** the robot in place at the end of the match without losing a token if you won't launch again.
 - Don't separate **Dual Lock**, break models, or combine models so they can't be freed in **one motion**.
 - Don't interfere with the **other team's field** (unless a mission says you can).
- **Tips for Kids:**
 - Program a **safe return** path after each mission to avoid token loss.

- **Examples:**
 - If a tool drops and rests **partly inside** home, you can remove it by hand—but if it's equipment, you'll **lose one token**.
- **Resources for Kids:** Interruption and handling rules.

Rules – After the Match: Scoring

- **Summary:** Time's up—hands off, and the referee scores what they see.
- **Key Points:**
 - When **2.5 minutes** ends, **stop** and **don't touch** anything.
 - Mission results must be **visible at the end** (unless the mission required a method).
 - Lines and **airspace** above an area count as **in** unless the mission says otherwise.
 - You and the referee agree on the score; if not, the **head referee** decides.
 - Only your **best of three** official matches counts; ties use 2nd and 3rd best.
- **Tips for Kids:**
 - Before time ends, make sure mission results are clearly **in position**.
- **Examples:**
 - If a model must be “in” a zone, even touching the **line** can count.
- **Resources for Kids:** Scoring procedure and tie rules.

Gracious Professionalism® (GP) at the Robot Game

- **Summary:** How you act matters; GP points affect Core Values scoring.
- **Key Points:**
 - Most teams earn **ACCOMPLISHED (3 pts)**.
 - **EXCEEDS (4 pts)** for above-and-beyond sportsmanship; **DEVELOPING (2 pts)** if sportsmanship needs work.
 - If you **don't show**, you get **0 GP**. If you show and explain issues, you can still earn **2–4**.
- **Tips for Kids:**
 - Help other teams, be kind to volunteers, and handle surprises calmly.
- **Examples:**
 - If the field is slightly different, adapt with a smile and ask politely for checks **before** the match.
- **Resources for Kids:** GP rubric summary.

Precision Tokens (Play Clean, Keep Points)

- **Summary:** You start with **6** tokens worth up to **50 points total**. You lose one when you interrupt the robot **outside home**.
- **Key Points (points if this many remain at the end):**
 - **1: 10 2: 15 3: 25 4: 35 5: 50 6: 50**
- **Tips for Kids:**
 - Plan missions to **avoid** touching the robot outside home.
- **Examples:**
 - Let the robot finish a run or return to home before you adjust anything.
- **Resources for Kids:** Token chart and references to rules on interruption.